Cardiac Pacemaker

Function, Components, and Clinical Applications

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1. Introduction

A cardiac pacemaker is a small, implantable medical device that delivers controlled electrical impulses to the heart muscle to regulate its rhythm. It is primarily indicated for patients suffering from bradyarrhythmias — conditions in which the heart beats too slowly to maintain adequate cardiac output. According to the World Health Organization, cardiovascular diseases remain the leading cause of mortality globally, and pacemakers represent a pivotal intervention in managing electrical conduction disorders. More than 1.2 million pacemakers are implanted worldwide each year, underscoring their critical role in modern cardiology (Mond & Proclemer, 2011).

The concept of artificial cardiac pacing dates to 1958, when Ake Senning and Rune Elmqvist implanted the first fully internal pacemaker in Stockholm, Sweden — a landmark in the history of medical technology (Jeffrey, 2001).

2. Function

The primary function of a pacemaker is to monitor the heart's intrinsic electrical activity and deliver timed electrical stimuli when the natural rate falls below a programmed threshold. Modern devices operate in demand mode: they remain passive when the heart beats normally and intervene only when needed.

Pacemakers are classified by chamber configuration:

- Single-chamber pacemaker: paces either the right atrium or right ventricle.
- Dual-chamber pacemaker: paces both the atrium and ventricle, preserving AV synchrony.
- Biventricular (CRT) pacemaker: coordinates both ventricles; used in heart failure with dyssynchrony.

3. Components

A pacemaker system consists of three main elements:

Component	Function
Battery (Lithium Iodide)	Powers device for 5–15 years
Leads (Electrodes)	Sense cardiac signals & deliver pulses
Pulse Generator	Controls timing & output of impulses
Microprocessor	Processes sensing data & adapts pacing

The pulse generator and battery are housed in a titanium casing (approximately 40–50 mm in diameter) implanted subcutaneously below the clavicle. Leads are flexible, insulated wires advanced transvenously into the cardiac chambers (Epstein et al., 2008).

4. Mechanism of Action

The sensing circuit continuously monitors intracardiac electrograms. If no intrinsic depolarization is detected within the programmed escape interval, the pulse generator delivers a low-amplitude impulse (typically 0.5–5 V, 0.4–1.0 ms duration) through the lead tip. This electrical stimulus initiates depolarization of adjacent myocardial cells, triggering a conducted heartbeat. The microprocessor adjusts parameters such as rate, output voltage, and sensitivity based on real-time data, enabling adaptive rate response with activity (Bernstein et al., 2002).

5. Clinical Applications

- Symptomatic Bradycardia: the most common indication — heart rate persistently below 60 bpm with associated symptoms (dizziness, syncope, fatigue).
- Sick Sinus Syndrome (SSS): dysfunction of the sinoatrial node causing inappropriate bradycardia or tachy-brady alternation.
- Atrioventricular (AV) Block (2nd and 3rd degree): failure of impulse conduction from atria to ventricles.
- Cardiac Resynchronization Therapy (CRT): biventricular pacing in heart failure patients with QRS prolongation to improve mechanical synchrony and ejection fraction (Abraham et al., 2002).

6. Risks & Complications

While pacemakers are generally safe, recognized complications include:

- Infection: pocket infection or lead endocarditis (~1–2% of implants); managed with antibiotics or device extraction.
- Lead dislodgement or fracture: may cause failure to sense or pace, requiring revision surgery.
- Pneumothorax: rare complication during subclavian venous access (~0.5%).
- Battery depletion: lithium-iodide batteries last 5–15 years; elective replacement is required before depletion (Epstein et al., 2008).
- Electromagnetic interference: strong magnetic fields (e.g., MRI) can interfere, although MRI-conditional devices are now available.

Pacemaker vs. Implantable Cardioverter-Defibrillator (ICD) — Quick Comparison

Feature	Pacemaker	Defibrillator (ICD)
Purpose	Treats slow heart rate	Treats fatal arrhythmias
Impulse type	Low-energy pulses	High-energy shocks
Main Condition	Bradycardia	VF / VT
Mode	Continuous/Demand	On-demand only

7. Conclusion

Cardiac pacemakers have transformed the management of bradyarrhythmias and conduction disorders, saving countless lives since their introduction in the late 1950s. Ongoing technological advances continue to expand their capabilities: the leadless pacemaker (e.g., Micra, Medtronic) — a capsule-sized device deployed entirely within the right ventricle — eliminates the need for a surgical pocket and transvenous leads, dramatically reducing infection risk (Reddy et al., 2015). Future directions include longer-lasting batteries, wireless remote monitoring, miniaturization, and AI-driven adaptive algorithms that predict arrhythmic events before they occur. As biophysics and bioengineering converge, the pacemaker will remain a cornerstone of life-saving cardiac therapy.

References

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